TABLE OF CONTENTS

Description - Contains theory for the CPU portion of the IPG Board. This theory is also applicable to IPG II Boards.

IPG 68020 CPU / VME Section

This section explains the operation of the section of the IPG Board entitled 68020 CPU/VME SECTION. There are eight blocks, each of which is discussed.

1- 68020 CPU - 68882 FPU BLOCK

A Motorola 68020 CPU device is used in conjunction with a 68882 floating point unit (FPU). The interface between them is simple, sharing data, address, and strobe lines. Selection of the coprocessor is accomplished through the application of the appropriate function code value (FC0-FC2 = 111 for CPU Space Cycle) and address. The size and A0 FPU inputs are configured for a 32-bit data bus width. The FPU sense output provides an indication that the FPU device is installed. PAL logic uses this signal to generate a bus error should an access be attempted to a missing FPU. The CPU Space Decoder PAL also provides the following other functions:

- Decodes the function code and address signals to generate IACK*
- Provides simple decoded read and write strobes (BDRD* and BDWR*)
- Provides an address strobe variant which is active for user and supervisory program and data spaces.

Reset logic gating is provided to allow the CPU to drive a reset signal out onto the board, through a software instruction, without resetting itself or its companion FPU.

2- EPROM BLOCK

A single 128 K x 16 bit EPROM device containing the boot code is provided for power-up. A jumper is provided to allow for expansion of this memory to 256 K x 16. The data bus is buffered in this block to decrease CPU loading. This buffered bus is used on other blocks as well.

3- DRAM BLOCK

A KS84C22 Dynamic Ram Controller (DRC) manages access and refreshes the four-mb array of dynamic RAM. The DRAM array is organized as four banks of 256 K x 32 bits. The DRAM Control PAL generates write enables and data buffer control signals for each of the four bytes individually to enable byte, word, or long word access as necessary. The DRC is programmed to interleave the operation of the banks to increase access time performance for sequential long word accesses. Staggered refresh operation (RAS only) is also programmed to reduce power supply noise.

The DRC is programmed for an active low early acknowledge output signal on DRAMACKIN*. This signal is further gated/delayed to ensure correct timing for VME sourced accesses.

4- INTERRUPT CONTROL BLOCK

Two 68901 multifunction peripheral chips are used on the IPG board. These provide a serial port, timers, general input/output pins, interrupt inputs, and interrupt prioritization and vector generation functions. One MFP device handles level-7 interrupts. The other MFP device handles level-6 interrupts. The two interrupt request signals are brought into the Interrupt Level Control PAL. Also brought into this PAL are the three priority encoded interrupt request lines from the VME bus interface device. The PAL then properly encodes the interrupt request, which is passed on to the CPU. The PAL also generates the appropriate interrupt acknowledge signal to enable either of the MFP devices, or the VMEchip, to drive its interrupt vector onto the buffered data bus. The buffered data bus output enable signal is generated by the PAL.

A global interrupt enable input to the PAL is provided by GPIO4 of the second MFP device. A level-7 specific interrupt enable signal is provided by GPIO7 of the first MFP device, which allows level seven interrupts to be re-enabled during level seven interrupt routines. Jumpers JP2,3 are used to configure the operating system or for other software needs. Jumpers JP4,5 are used to configure timer prescaler inputs. An RS-232 interface device is provided for a serial debug port.

5- A/N DISPLAY BLOCK

An eight-character alphanumeric display is provided. The HDSP-2111 is a smart device that contains character memory, ASCII decoder, multiplexed LED scanning, cursor control, character blanking and blinking, and user-definable characters. The device, both readable and writable, is connected to the buffered data bus. A one-shot device generates a one-quarter-second flash on an LED each time a high to low transition on the bus grant acknowledge signal occurs. This indicates an access to local resources from the VME bus.

6- DIAGNOSTICS CIRCUITS BLOCK

A MAXPAL device is central to the diagnostic circuit block. It contains the following functions:

- A 24-bit sequence timer, with overflow bit, that counts microseconds from the last SSI* interrupt. This same timer ensures the minimum length of time for each output reset pulse (128 msec).
- A 32-bit test register: writable/inverse readable, the lower two bits of which also drive test LEDs
- A 32-bit DRC configuration value
- Power up and warm start status flip flops
- An eight-bit register containing reset bits for the SPI, SPU, and WARP functions of the IPG. In addition, reset bits for off board resources of SRI and MDS are provided. The mode load bit for the DRC and a SCORPIO self-reset bit complete the list.

The reset logic of the IPG board is somewhat complex. A reset input may be presented from a number of sources, three of which are:

- An IPG front panel push button
- An event on the +5 VDC supply
- Activation of the SCORPIO bit

Any of these events generates an active low signal on the BDRESET* input to the MAXPAL. Inside the MAXPAL, a flip flop is cleared after 300 ns, causing the MAXPAL output BDRESOUT* to become active. This signal is input to the VME bus interface device, where it is logically ANDed with the SYSRESET* signal from the backplane itself. The result is a signal called PRESET*, which is the actual reset signal used by much of the board. When one of the three previously identified reset sources is removed, BDRESET* becomes inactive, releasing the clear input from the MAXPAL flip flop. Inside the MAXPAL, the 24-bit counter is also released to count 1 mHz clock pulses. When bit 16 of this countertransitions from low to high, the flip flop is clocked with a value of 1 applied to its D input. This drives the flip flop output BDRESOUT* inactive once again, which in turn causes PRESET* to also go inactive. As noted earlier, the backplane reset input simply passes through the VMEchip without further gating.

7- BUS BUFFER BLOCK

Bus transceivers simply buffer the 32-bit data bus before routing these signals across the PWB. Similarly, 24 bits of the address bus are buffered for increased drive. A LOCAL BUS ADDRESS DECODE PAL is provided to generate memory segment enables which are used in the PULSE SEQUENCER AND WARP PROCESSOR sections of the board.

8- VME BUS I/F BLOCK

The main functionality is provided by a VME Bus interface gate array, the MVME6000. It is directly connected to the VME Bus, including the lower eight bits of the address and data busses. It provides buffer direction, latch clock, and output enable control signals for 74F543 transceivers used to buffer the remaining 24 bits of the address and data buffers. In addition, control is provided for data buffers to the local data bus, which are capable of word swapping.

The VME chip is a programmable device that allows software to configure such things as address modifier codes used/accepted, watchdog time constants, etc. A group of global registers is provided for interprocessor communications.

Two PALs are used in the block called ADDRESS_DECODER to decode the standard and extended address regions to which the IPG board may respond, provided the appropriate address modifier code is also present. The PALs allow access to the local DRAM, waveform memory, instruction memory, and SPU and SPI memory.

The generation of DSACKs meets VME timing on the board for all but the DRAM array. As described in the DRAM section, the DRC provides an early acknowledge. This acknowledge must be delayed for VME accesses to DRAM to avoid a protocol crash. This is also accomplished by the BUS GRANT CONTROL PAL.

To ensure correct operation of the word swap buffers when sequencer and WARP processor memories are accessed, the MVME6000 swap buffer control signals are modified by PAL.

9- LOCAL ADDRESS DECODER BLOCK

Two PALs are used for decoding, and one generates the appropriate combination of DSACK signals, adding the necessary number of wait states. Fast NOR gates are used to reduce the upper six address bits to a pair before input to the PALs, maximizing the available pin out. Due to loading on the DSACK lines, a bipolar PAL device is necessary.

10- CLOCK GENERATOR BLOCK

The IPG board was designed to operate from a single clock source, which is a 20 MHZ Exciter clock available on the backplane. This was done to prevent various clocks of an asynchronous nature from generating noise which would be deleterious to the system image quality. Still multiple clock frequencies are required on the board. These include 33 mHz for the SPU and WARP processors, 11 mHz for the SPI, and 10 and 5 mHz for the sequencers. To develop all of these frequencies, the 20-mHz clock is multiplied by five times to get 100 mHz, which is then divided by 3, 5, 9, 10, 20, and 40.

A DP8530 clock generator device provides a phase locked loop system to perform the frequency multiplication. A varactor diode tuned tank circuit is controlled by the lock error voltage to maintain phase lock with the 20-mHZzinput.

The 100-mHz ECL clock out is divided by 3, 5, and 9 with circuits including counters, delay lines, and OR gates. A 50% duty cycle is created. These clocks are translated back to TTL levels, and are multiply buffered for distribution across the board.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 2, 1998	J. Saperstein	Initial Conversion from Toolbook to Word.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.